BADR UNIVERSITY IN CAIRO (BUC)
Faculty of Engineering & Technology — Mechatronics & Energy Engineering

Course: Data Acquisition & LabVIEW (MCH 232)
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COMPREHENSIVE OFFICIAL SOLUTIONS GUIDE — SHEETS (1), (2), (3) & (4)

SHEET (1) : MEASUREMENT SYSTEMS, ARCHITECTURE & ADC PRINCIPLES

Course Code: MCH232

1) Draw and Discuss the Measurement System and Its Architecture. — Theory & Architecture

A. Architecture Block Diagram:



B. Discussion of Subsystems:

- **1. Primary Sensing Element (Sensor / Transducer):** Interacts directly with the physical measurand (e.g., temperature, displacement, pressure) and converts it into an analog electrical signal (voltage, current, resistance).
- **2. Signal Conditioning Stage:** Prepares the raw electrical signal for digitization through: *Amplification* (boosting weak mV signals), *Filtering* (anti-aliasing and noise attenuation), *Buffering* (high-impedance matching to eliminate sensor loading errors), and *Linearization*.
- **3. Analog-to-Digital Converter (ADC):** Samples the continuous analog voltage at discrete intervals (f_s), quantizes it into discrete amplitude bins, and outputs N-bit binary words.
- **4. Processing & Control Unit (PC / LabVIEW):** Executes mathematical operations, Fourier Transforms (FFT), data logging, and closed-loop control algorithms (PID).
- **5. Display / Storage / Output:** Renders graphs on the Front Panel, writes records to disk, and sends control commands to physical actuators.

2) Discuss the Characteristics of Measurement Systems. — Static vs Dynamic Characteristics

Measurement system performance is classified into two fundamental domains:

A. Static Characteristics (Steady-state or slowly varying inputs):

- **Accuracy:** Closeness of the measured value to the true physical value. Quantified by absolute or percentage error.
- **Precision (Repeatability):** The degree of mutual agreement among repeated measurements of the same quantity under identical conditions.
- **Sensitivity (S):** Ratio of change in output to change in input ($S = \Delta y / \Delta x$, the slope of the calibration curve).
- **Resolution:** The smallest increment in input that produces an observable and detectable change in output.
- **Linearity:** The maximum deviation of the actual calibration curve from a specified straight line.
- **Range & Span:** Range defines limits [x_{min} to x_{max}]; Span is dynamic width ($Span = x_{max} - x_{min}$).
- **Hysteresis:** The difference in output for the same input value when approached from an increasing direction versus a decreasing direction.

B. Dynamic Characteristics (Rapidly time-varying inputs):

- **Zero-Order Systems:** Output follows input instantaneously without phase lag ($y(t) = K \cdot x(t)$, e.g., potentiometer).
- **First-Order Systems:** Characterized by time constant τ ($\tau\ dy/dt + y = K\ x(t)$, e.g., thermocouple, RC filter). Output reaches 63.2% of step response at $t = \tau$.
- **Second-Order Systems:** Governed by natural frequency ω_n and damping ratio ζ ($d^2y/dt^2 + 2\zeta\omega_n\ dy/dt + \omega_n^2 y = K\ \omega_n^2\ x(t)$, e.g., accelerometers, pressure transducers). Characterized by resonance, overshoot, and bandwidth.

3) In the Given Photo, Describe Each Component and Its Job. — Internal DAQ Hardware Architecture



Component	Technical Role & Exact Engineering Job
1. Inputs	Terminal interfaces receiving physical analog voltages from multiple sensors (thermocouples, pressure sensors, strain gauges).
2. Buffer (Impedance Matcher)	A high-input-impedance unity-gain operational amplifier (voltage follower). It prevents **loading effects** on delicate sensors, ensuring voltage is transferred without attenuation.
3. Gain and Filtering	• **Programmable Gain Amplifier (PGA):** Scales weak signals up to match the ADC's full input span, maximizing digital resolution. • **Anti-Aliasing Filter:** Analog low-pass filter cutting off frequencies above f_s / 2 to eliminate aliasing and high-frequency noise.
4. Sample and Hold (S/H)	Captures instantaneous signal voltage at precise clock moments (*Sample mode*) and stores it across an internal capacitor (*Hold mode*). Keeps voltage constant during ADC conversion time, eliminating **aperture jitter error**.
5. A/D Converter (ADC)	Quantizes the held analog voltage into discrete amplitude steps and encodes it into an N-bit binary digital word (2^N discrete levels).
6. Multiplexer (MUX)	Selectively routes digital output words or switches between multiple input channels, allowing multiple sensor channels to share a single processing bus or ADC.

4) Compare Between Analog Signal and Digital Signal with Examples. — Comparison Table

Comparison Feature	Analog Signal	Digital Signal
Definition	Continuous in both time and amplitude. Can assume an infinite number of values within a range.	Discrete in time and quantized in amplitude. Represented by binary codes (0s and 1s).
Waveform	Smooth continuous curves (sine, triangular, real-world physical functions).	Square pulses, binary logic level sequences (0V / 5V).
Noise Immunity	Poor. Highly vulnerable to noise, EMI, attenuation, and cumulative distortion.	Very high. Logic levels are immune to moderate noise. Error checking and parity can be applied.
Processing & Storage	Processed via analog hardware (op-amps, passive filters). Hard to store without loss.	Easily processed by microprocessors, DSPs, and LabVIEW. Stored indefinitely without degradation.
Real-World Examples	• Voltage output from a Thermocouple (µV to mV). • Acoustic pressure waves picked up by a microphone. • Fluid pressure diaphragm analog output (4–20 mA).	• 16-bit binary output word from a DAQ board. • Digital pulses from an optical rotary shaft encoder. • CAN bus / USB communication packets between ECU and PC.

5) What is the Purpose of ADC and How Does It Work? — Operational Principle of ADC

A. Primary Purpose:
Physical quantities exist in continuous analog form. Microprocessors, computers, and software (such as LabVIEW) can only understand discrete binary logic. The **ADC (Analog-to-Digital Converter)** acts as the fundamental bridge, converting continuous analog voltages into equivalent discrete N-bit digital binary numbers.

B. The 3 Sequential Steps of ADC Operation:

1. **1. Sampling (Time Discretization):** The continuous input voltage $V(t)$ is sampled at periodic time intervals $T_s = 1 / f_s$. The Sample & Hold circuit freezes this voltage during conversion.
2. **2. Quantization (Amplitude Discretization):** The total input voltage range ($Span = V_{max} - V_{min}$) is partitioned into 2^N discrete vertical bins separated by step size:
$$Q = (V_{max} - V_{min}) / 2^N$$
The instantaneous sampled voltage is assigned to the nearest quantization bin.
3. **3. Encoding (Binary Word Generation):** The assigned quantization level index (integer $D \in [0, 2^N - 1]$) is converted into an N-bit binary word ($b_{N-1}\ b_{N-2} \ldots b_0$).

6) How Can I Select an ADC Which Will Be the Best Choice for My Measurement System?

The 6-Point Selection Criteria

> **Engineering Selection Principle:** The best ADC is the one that covers the entire signal span with **Zero Clipping** while delivering the **Finest Resolution (Smallest Q)** at an adequate sampling speed.

- **1. Input Voltage Range & Polarity (No Clipping Filter):** Must safely match the sensor's extreme voltages ($V_{min} \leq S_{min}$ and $V_{max} \geq S_{max}$). If the ADC range is narrower than the signal, clipping occurs, destroying physical data.
- **2. Number of Bits / Resolution (N):** Determines the smallest detectable voltage step $Q = Span / 2^N$. Higher bit count reduces quantization error ($e_q = \pm Q / 2$).
- **3. Maximum Sampling Rate (f_s):** Must satisfy Nyquist theorem ($f_s \geq 2\ f_{max}$) to prevent aliasing. For good waveform fidelity, choose $f_s \geq 5$ to $10\ f_{max}$.
- **4. Channel Count & Architecture:** Single-ended vs. Differential (differential rejects common-mode noise). Simultaneous sampling ADCs for phase matching vs. Multiplexed ADCs for lower cost.
- **5. Dynamic Range & ENOB:** Effective Number of Bits (ENOB) accounts for internal noise, distortion, and non-linearity (INL/DNL).
- **6. Interface & Bus Bandwidth:** USB, PCIe, or Ethernet throughput to ensure continuous, gap-free streaming into LabVIEW memory without buffer overrun.

SHEET (2) : 16-BIT LAB DAQ SYSTEM & FFT FREQUENCY SPECTRUM

Questions 1 to 4

Questions 1 & 2: 16-Bit ADC Output Levels, Overall Resolution & Quantization Error

Given: 16-bit ADC, Input Range [-10 V to +10 V]

Governing Equations:

- Span: $\text{Span} = V_{max} - V_{min} = 10\ V - (-10\ V) = 20\ V$
- Number of Levels (Bins): $\text{Levels} = 2^N = 2^{16}$
- Overall Resolution (Q): $Q = \text{Span} / 2^N = (V_{max} - V_{min}) / 2^N$
- Quantization Error (± Resolution): $e_q = \pm Q / 2$

Solution 1: Number of Available Bins / Levels [2 Marks]

For an $N = 16$-bit ADC:

$$\textbf{Available Bins / Levels} = 2^{16} = \textbf{65,536 levels}$$

Solution 2: Overall Resolution & ± Resolution (to 3 Significant Digits) [2 Marks]

- **Overall Resolution (Q):**
  $Q = 20\ V / 65536 = 0.00030517578\ldots\ V = 0.305\ mV$
- **± Resolution (Quantization Error e_q):**
  $e_q = \pm Q / 2 = \pm (0.30517578\ mV / 2) = \pm\ 0.153\ mV$

Conclusion: The system provides **65,536 levels**, an overall step resolution of **0.305 mV**, and a quantization uncertainty of **±0.153 mV**.

Question 3: Effective Bit Utilization for Unipolar Signal [0 to 2.5 V]

Given: Input Range [-10 V to +10 V], Signal [0 V to 2.5 V]

Analysis: Total ADC Span = 20 V. Signal Span = 2.5 V - 0 V = 2.5 V.

Step 1: Calculate Fraction of ADC Range Utilized:

$$\textbf{Fraction Utilized} = \textbf{Signal Span / Total ADC Span} = \textbf{2.5 V / 20 V} = \textbf{1/8} = \textbf{2}^{-3}$$

Step 2: Number of Quantization Levels Utilized:

$$N_{\text{levels, eff}} = 65{,}536 \times (1/8) = \textbf{8,192 levels}$$

Step 3: Effective Number of Bits Utilized (N_{eff}):

$$2^{N_{eff}} = 8{,}192 \Rightarrow N_{eff} = \log_2(8192) = \textbf{13 bits}$$

Engineering Insight: Exactly **3 bits are wasted** (16 - 13 = 3 bits):

- **1 bit** is wasted because the signal is purely positive (wasting the entire [-10V, 0V] negative half).
- **2 bits** are wasted because the signal reaches only +2.5 V out of the +10 V positive span ($2.5 / 10 = 1/4 = 2^{-2}$).

Question 4: FFT Parameters for 50 Hz Sine Wave ($N = 256$, $f_s = 200$ Hz) — Total Time, Frequency Resolution & Max Useful Frequency

> **FFT Relations:** Total Sampling Time $T = N / f_s$ | Frequency Resolution $\Delta f = f_s / N = 1 / T$ | Max Useful Frequency $f_{max,useful} = f_s / 2$

- **1. Total Sampling Time (T):**
  $T = N / f_s = 256 / 200 \text{ Hz} = 1.28 \text{ seconds}$
- **2. Frequency Resolution (Δf or Bin Width):**
  $\Delta f = f_s / N = 200 \text{ Hz} / 256 = 1 / 1.28 \text{ s} = 0.78125 \text{ Hz} \approx 0.781 \text{ Hz}$
- **3. Maximum Useful Frequency ($f_{max,useful}$):**
  By Nyquist criterion, the FFT spectrum is symmetric about the folding frequency. Frequencies above $f_s / 2$ represent redundant reflections.
  $f_{max,useful} = f_s / 2 = 200 \text{ Hz} / 2 = 100 \text{ Hz}$

> **Spectral Verification:** The 50 Hz signal falls cleanly at bin index $k = 50 / 0.78125 = 64$, well within the useful range [0, 100 Hz].

SHEET (3) : MOTOR VIBRATION, RPM & NYQUIST SAMPLING

Questions 5 and 6

Question 5: Minimum Sampling Frequency to Measure Motor RPM (Max RPM = 1800) [4 Marks]

Nyquist Rate Calculation

Fundamental Physics: A vibrating motor shaft rotating at RPM completes RPM / 60 revolutions per second (Hz).

Step 1: Convert Maximum RPM to Rotational Frequency (f_{max} in Hz):

$$f_{max} = RPM_{max} / 60 = 1800 / 60 = 30 \text{ Hz}$$

Step 2: Apply the Nyquist Sampling Rate Theorem:

To avoid aliasing and uniquely reconstruct a periodic signal, the sampling frequency f_s must be at least twice the highest frequency component of the signal:

$$f_s \geq 2 \cdot f_{max} \Rightarrow f_{s,min} = 2 \times 30 \text{ Hz} = 60 \text{ Hz}$$

Explanation for Full Credit: At $f_s = 60$ Hz, exactly 2 samples are acquired per revolution of the motor shaft (one near the wave's peak and one near the trough). This is the absolute minimum mathematical threshold required to detect the 30 Hz fundamental vibration frequency without aliasing.

Question 6: Minimum Number of Data Points for 2 RPM Resolution [2 Marks]

Frequency Resolution & Record Length

Governing Equations:

- Required Frequency Resolution: $\Delta f = \Delta RPM / 60$
- Required Total Acquisition Time: $T = 1 / \Delta f$
- Minimum Number of Discrete Sample Points: $N = f_s \cdot T$

Step 1: Convert Required RPM Resolution to Frequency Resolution (Δf):

$$\Delta f = \Delta RPM / 60 = 2 \text{ RPM} / 60 = 1/30 \text{ Hz} \approx 0.03333 \text{ Hz}$$

Step 2: Determine Required Total Sampling Duration (T):

$$T = 1 / \Delta f = 1 / (1/30 \text{ Hz}) = 30 \text{ seconds}$$

Step 3: Calculate Minimum Number of Points (N) at $f_s = 60$ Hz:

$$N = f_{s,min} \times T = 60 \text{ Hz} \times 30 \text{ s} = 1800 \text{ points}$$

Direct Identity Verification:

$N = f_s \cdot T = (2 f) \cdot (1 / \Delta f) = (2 \cdot RPM / 60) \cdot (60 / \Delta RPM) = 2 \cdot (RPM / \Delta RPM) = 2 \cdot (1800 / 2) = 1800$ points

SHEET (4) : DISCRETE SAMPLING, ALIASING & FOLDING DIAGRAM

Scenarios (a, b, c, d) and Problem 6

Question 1: Calculate Perceived Frequency (f_a) Inferred from Acquired Data

Scenarios (a), (b), (c), and (d) with Folding Diagram

Universal Governing Equations:

• Folding Frequency: $f_{folding} = f_N = f_s / 2$ | • Cimbala Universal Formula: $f_a = | f - n \cdot f_s |$ where $n = \text{round}(f / f_s)$

Scenario	Signal f	Sampling f_s	$f_N = f_s / 2$	Sampling Regime & Formula	Perceived Frequency (f_a)
(a)	800 Hz	2000 Hz	1000 Hz	**Regime 1:** $f_s \geq 2f$ (2000 ≥ 1600 Hz). Nyquist satisfied → **No Aliasing!**	**fa = 800 Hz**
(b)	700 Hz	900 Hz	450 Hz	**Regime 2:** ⅔ f < f_s < 2f (466.7 < 900 < 1400 Hz). Simple formula: $f_a =	f_s - f	=	900 - 700 |$	**fa = 200 Hz**
(c)	1000 Hz	1000 Hz	500 Hz	**Aliasing:** $n = \text{round}(1000/1000) = 1$. $f_a = | 1000 - 1 \times 1000 | = 0$ Hz. (All samples at same phase)	**fa = 0 Hz (DC)**
(d)	600 Hz	500 Hz	250 Hz	**Regime 2:** ⅔ f < f_s < 2f (400 < 500 < 1200 Hz). Simple formula: $f_a =	f_s - f	=	500 - 600 |$	**fa = 100 Hz**

Verification via Folding Diagram for Each Scenario:

- **(a)** Ratio $f / f_N = 800 / 1000 = 0.8$ → Lies on baseline before folding → $f_a = 0.8 \times 1000 = 800$ Hz.
- **(b)** Ratio $f / f_N = 700 / 450 = 1.556$ → Lies on Segment 1 ($X = 2 - 1.556 = 0.444$) → $f_a = 0.444 \times 450 = 200$ Hz.
- **(c)** Ratio $f / f_N = 1000 / 500 = 2.0$ → Lies on corner 2.0 at horizontal $X = 0.0$ → $f_a = 0.0 \times 500 = 0$ Hz.
- **(d)** Ratio $f / f_N = 600 / 250 = 2.40$ → Lies on Segment 2 ($X = 2.40 - 2.0 = 0.40$) → $f_a = 0.40 \times 250 = 100$ Hz.

Problem 6: Voltage Signal $f = 600\ Hz$, Sampled at $f_s = 320\ Hz$ **Severe Undersampling Trap**

> ⚠️ **FATAL EXAM TRAP (Regime 3 Check):**
> Check two-thirds frequency limit: $\frac{2}{3} f = \frac{2}{3} (600\ Hz) = 400\ Hz$.
> Since $f_s = 320\ Hz \leq \frac{2}{3} f$, this is **Severe Undersampling (Regime 3)**!
> ❌ **NEVER use the simple formula:** $| f_s - f | = | 320 - 600 | = 280\ Hz$ ← **COMPLETELY WRONG!**

Method 1: Cimbala Universal Formula (Recommended)

1. Calculate ratio and nearest integer n:

$$f / f_s = 600 / 320 = 1.875 \Rightarrow n = \text{round}(1.875) = 2$$

2. Substitute into universal equation:

$$f_a = | f - n \cdot f_s | = | 600 - 2 \times 320 | = | 600 - 640 | = |-40| = 40\ Hz$$

Method 2: Normalized Folding Diagram

1. Calculate folding frequency: $f_{folding} = f_N = f_s / 2 = 320 / 2 = 160\ Hz$.
2. Calculate dimensionless ratio: $f / f_{folding} = 600 / 160 = 3.75$.
3. Locate 3.75 on the folding diagram: It falls on **Segment 3** (between 3.0 and 4.0), which has the governing line equation $X = 4.0 - Y$:

$$f_a / f_{folding} = 4.0 - 3.75 = 0.25$$

4. Compute perceived frequency:

$$f_a = 0.25 \times f_{folding} = 0.25 \times 160\ Hz = 40\ Hz$$

> **Final Verified Result:** The perceived frequency displayed by the digital DAQ system is exactly **fa = 40 Hz**!